PROJECT PROPOSAL

MAY 2026 SEMESTER

ChiroMemo – Personal Learning and Knowledge Management System (PLKMS)

BY

MUHAMMAD HANIF IBRAHIM

SCHOOL OF SCIENCE AND TECHNOLOGY

ASIA e UNIVERSITY

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By :
MUHAMMAD HANIF IBRAHIM
C30109230009

SCHOOL OF SCIENCE AND TECHNOLOGY

ASIA e UNIVERSITY

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C30109230009

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ABSTRACT

The increasing use of digital learning resources has transformed how university students acquire knowledge through online courses, documents, websites, videos, and personal notes. However, these resources are often distributed across multiple platforms, making it difficult to organize information, track learning progress, and manage knowledge effectively. This project proposes the development of a Personal Learning and Knowledge Management System (PLKMS), a web-based application designed to support self-directed learning by integrating learning management and knowledge organization into a single platform. The proposed system provides features for managing learning goals, organizing learning resources, tracking progress, and supporting knowledge management through AI-assisted functionalities such as intelligent search, document summarization, and personalized learning recommendations. The project adopts the Object-Oriented Analysis and Design (OOAD) approach together with the Agile software development methodology. The expected outcome is an integrated platform that enables university students to organize learning materials more effectively, monitor their academic progress, improve knowledge retention, and enhance their overall self-directed learning experience.

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CHAPTER 1
INTRODUCTION

1.1 Overview

The rapid growth of digital technology has significantly transformed the way people access and acquire knowledge. Today, information can be obtained from various sources such as online courses, educational websites, videos, e-books, research articles, and personal notes. This accessibility provides learners with greater opportunities to expand their knowledge and develop new skills independently. For university students, access to diverse learning resources is particularly important in supporting academic performance and lifelong learning.

Despite these advantages, the abundance of learning resources often creates challenges in managing and organizing information effectively. Students frequently rely on multiple platforms to store notes, learning materials, and progress records, resulting in fragmented information and inefficient knowledge management. As a consequence, learners may experience difficulties in locating previously studied materials, tracking their learning progress, and maintaining a structured learning pathway.

To address these challenges, this project proposes the development of ChiroMemo, a Personal Learning and Knowledge Management System (PLKMS) designed to assist university students in organizing learning resources, managing personal knowledge, monitoring learning progress, and supporting self-directed learning through an integrated web-based platform.

1.2 Problem Statement

The increasing availability of digital learning resources has created several challenges for university students in managing their learning activities effectively.

First, learning materials are often scattered across multiple platforms such as online learning websites, cloud storage services, note-taking applications, and social media. This fragmentation makes it difficult for students to organize, manage, and retrieve learning information efficiently.

Second, many students struggle to monitor their learning progress and maintain a structured learning plan. Without a centralized system to track learning goals, completed activities, and learning achievements, students may lose focus and experience difficulties in prioritizing their studies.

Third, existing learning tools generally focus on specific functions, such as note-taking, task management, learning management, or content storage. As a result, students may need to use multiple applications simultaneously, which can lead to fragmented information and an inefficient learning experience.

Therefore, there is a need for an integrated system that combines learning goal management, learning roadmap management, learning resource organization, personal knowledge management, progress tracking, and AI-assisted learning support within a single platform.

1.3 Objectives

The objectives of this project are:

1. To develop a web-based Personal Learning and Knowledge Management System (PLKMS) called ChiroMemo that enables university students to organize and manage their learning resources within a centralized platform.
2. To implement AI-assisted features, including intelligent search, document summarization, and personalized learning recommendations, to support students in managing and accessing learning knowledge effectively.
3. To provide learning goal management, learning roadmap management, and progress tracking functionalities that help university students monitor their learning activities and achieve their learning goals more effectively.

1.4 Scope

The scope of this project is defined as follows:

Functional Scope

- User registration and login.
- Learning goal management.
- Learning roadmap management.
- Learning progress tracking.
- Learning resource management.
- Personal note management.
- AI-powered intelligent search.
- AI-powered document summarization.
- Personalized learning recommendations.
- Dashboard for monitoring learning activities.

Technical Scope

- The system is developed as a web-based application.

- The system is accessible through modern web browsers.
- Internet connectivity is required to access AI-powered features.

Project Limitations

- The system is intended for university students only.
- The system does not support lecturer or administrator modules.
- Collaborative learning features are not included.
- The system is designed for personal learning and knowledge management only.

1.5 Target User

The target users of the proposed system are:

- Primary User: University students.
- Students who utilize multiple digital learning resources, such as online courses, educational websites, research papers, videos, and personal notes.
- Students who need a centralized platform to organize learning materials, monitor learning progress, and manage personal knowledge.
- Students who practice self-directed learning and wish to improve their learning efficiency through AI-assisted features.

1.6 Methodology

This project adopts the Object-Oriented Analysis and Design (OOAD) approach together with the Agile software development methodology to guide the development of the proposed system. OOAD is employed to analyze user requirements and design the system using object-oriented principles, resulting in a structured, modular, and maintainable system architecture.

The Agile methodology is selected because it promotes an iterative and incremental development process, allowing continuous refinement based on project progress and user feedback. Through Agile, the system will be developed in several development cycles, enabling features to be implemented, tested, and improved progressively. The combination of OOAD and Agile provides a systematic and flexible approach for developing ChiroMemo while ensuring that the final system meets the project objectives and user requirements.

1.7 Project Timeline

No.	Activity	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11-12	Week 13-14	Week 15-16
1	Literature Review	●	●						
2	Requirement Analysis		●	●					
3	System Design (DFD, ERD, Flowchart, UI)			●	●				
4	Database Development				●	●			
5	Frontend Development				●	●	●		
6	Backend Development				●	●	●		
7	AI Feature Integration					●	●		
8	System Testing & Debugging						●	●	
9	Documentation & Report Writing	●	●	●	●	●	●	●	●
10	Final Review & Project Submission							●	●

1.1 Project Timeline

CHAPTER 2
LITERATURE REVIEW

1.1 Introduction

Advances in digital technology have significantly transformed the way students acquire, manage, and utilize information in the learning process. The use of digital technology enables quick and flexible access to a variety of learning resources, thereby supporting a more independent and learner-centered learning process (Bond et al., 2020). However, this ease of access also presents a challenge in the form of an increasing volume of information that must be selected, organized, and utilized effectively in order to become meaningful knowledge (Siemens, 2005).

As the concept of digital learning evolves, students are expected not only to acquire information but also to systematically manage the knowledge they possess. Effective knowledge management can help enhance learning effectiveness, support decision-making, and facilitate lifelong learning (Davenport & Prusak, 1998). Furthermore, advancements in Artificial Intelligence (AI) open new opportunities to support personalized learning through the provision of recommendations, intelligent information retrieval, and automatic content summarization (Holmes et al., 2022).

This chapter discusses the theories that form the foundation for the development of the proposed Personal Learning and Knowledge Management System. The discussion covers the concepts of Digital Learning, Personal Learning, Knowledge Management, Personal Knowledge Management, Self-Directed Learning, Artificial Intelligence, and the proposed system. In addition, this chapter also reviews several widely used systems—namely Notion, Obsidian, and Moodle—as a basis for identifying the strengths, limitations, and opportunities for developing the proposed system

2.1.1 Digital Learning

Digital learning is an educational approach that utilizes digital technology to support the delivery of course material, communication, collaboration, and learning assessment. According to Bond et al. (2020), digital learning enables students to have a more flexible learning experience because learning activities can be conducted without being limited by time or place through various digital devices.

The development of digital learning has given rise to various learning platforms, such as Learning Management Systems (LMS), Massive Open Online Courses (MOOCs), educational videos, mobile learning, and Artificial Intelligence-based applications. These technologies make it easier for students to access various learning resources but also increase the risk of information overload—a condition where users receive such a vast amount of information that they struggle to select relevant information (Bawden & Robinson, 2009).

In higher education, digital learning serves not only as a medium for delivering course content but also supports collaborative learning, self-directed learning, and the development of 21st-century skills, such as critical thinking, problem-solving, communication, and digital literacy (Bond et al., 2020). Therefore, a system is needed that can help students organize various learning resources so that the information they obtain can be transformed into useful knowledge.

2.1.2 Personal Learning

Personal learning is a learning approach that places the learner at the center of the entire learning process. In this approach, individuals have the freedom to set learning goals, choose learning resources, develop learning strategies, and evaluate their progress based on their own needs and characteristics (Attwell, 2007).

The concept of personal learning has evolved alongside the emergence of the Personal Learning Environment (PLE)—a learning environment built by individuals using various digital applications and technologies to support self-directed learning (Attwell, 2007). A PLE enables students to integrate diverse information sources, note-taking apps, learning media, and collaboration platforms into a single learning ecosystem tailored to their needs.

The implementation of personal learning is expected to enhance students' motivation, independence, and ability to develop competencies aligned with their interests and desired learning outcomes. Therefore, this concept serves as one of the main foundations for the development of the proposed system.

2.1.3 Knowledge Management

Knowledge Management (KM) is a systematic process of creating, acquiring, storing, sharing, and utilizing knowledge to enhance the effectiveness of both individuals and organizations. According to Davenport and Prusak (1998), knowledge management focuses not only on managing information but also on how that information is transformed into knowledge that can be used to support decision-making and problem-solving. Thus, knowledge management serves as a mechanism that helps individuals derive maximum benefit from the knowledge they possess.

In the digital age, students obtain information from various sources, such as scientific journals, e-books, instructional videos, discussion forums, and online learning platforms. Without a system capable of managing this information in a structured manner, students often struggle to retrieve material they have previously studied, thereby making the learning process less effective.

Based on this concept, this study applies the principles of knowledge management in the development of a Personal Learning and Knowledge Management System (PLKMS). The proposed system provides facilities for storing learning materials, managing learning resources, recording information, and systematically organizing knowledge so that students can acquire, store, and utilize knowledge more effectively.

2.1.4 Personal Knowledge Management

Personal Knowledge Management (PKM) is the application of knowledge management concepts at the individual level, focusing on a person's ability to manage information and knowledge to support learning and work activities. Avery et al. (2001) explain that personal knowledge management is a series of activities that include collecting, organizing, storing, retrieving, and reusing an individual's information so that it can be used effectively.

In the context of higher education, students often use various different applications to store notes, references, study schedules, and academic assignments. This situation results in information being scattered across various platforms, making it difficult to retrieve when needed.

Therefore, this study developed PLKMS as a system that integrates various personal knowledge management activities into a single platform. The system provides features for note-taking, learning resource management, grouping information by topic, and smart search capabilities so that students can manage their personal knowledge in a more structured and efficient manner.

2.1.5 Self-Directed Learning

Self-Directed Learning (SDL) is a learning approach that places the individual in charge of their entire learning process. Knowles (1975) defines self-directed learning as the process by which individuals independently identify their learning needs, set learning goals, select learning strategies, utilize learning resources, and evaluate the learning outcomes they have achieved.

Self-directed learning skills are becoming increasingly important in the digital age because students have access to a vast array of information sources. However, the success of self-directed learning depends not only on the availability of information but also on students' ability to manage their time, maintain consistency in their learning, and organize the knowledge they acquire.

In this study, the concept of self-directed learning is implemented through the Learning Goals, Learning Roadmap, Progress Tracking, and learning activity reminders features. Through these features, students can set learning targets, create study plans, monitor their progress, and evaluate their achievements independently, thereby making the learning process more focused.

2.1.6 Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on developing systems capable of performing tasks that typically require human intelligence, such as understanding language, recognizing patterns, reasoning, and making decisions. Russell and Norvig (2021) explain that the primary goal of AI is to develop intelligent agents capable of observing their environment, processing information, and taking actions that yield the best results based on specific objectives.

Advances in AI have brought significant changes to the field of education. Zawacki-Richter et al. (2019) note that the application of AI in education includes intelligent tutoring systems, adaptive learning, automated assessment, learning analytics, chatbots, and recommendation systems. These technologies enable a more personalized learning process, as the systems can tailor their services based on users' characteristics and needs.

In this study, AI is utilized to help students search for information intelligently, generate summaries of learning materials, and provide recommendations for relevant learning resources based on the topics they are studying. Thus, AI is expected to improve the efficiency of the learning process while reducing the burden on students in managing information.

2.1.7 Proposed System

The system proposed in this study is a Personal Learning and Knowledge Management System specifically designed to help students manage their learning and knowledge in an integrated manner.

This system combines the concepts of personal learning, knowledge management, self-directed learning, and artificial intelligence into a single platform. Some of the key features include learning goal management, learning roadmaps, learning resource storage, note-taking, learning progress tracking, AI-powered smart search, automatic content summarization, and learning resource recommendations.

By integrating these various functions, the system is expected to enhance the effectiveness of students' learning while helping them build a personal knowledge base that can be utilized in the long term.

2.2 Review on Existing System

Before developing the proposed system, it is important to review several platforms that have been widely used to support learning and knowledge management. This analysis aims to identify the strengths, weaknesses, and opportunities for development that can be applied to the proposed system.

2.2.1 Notion

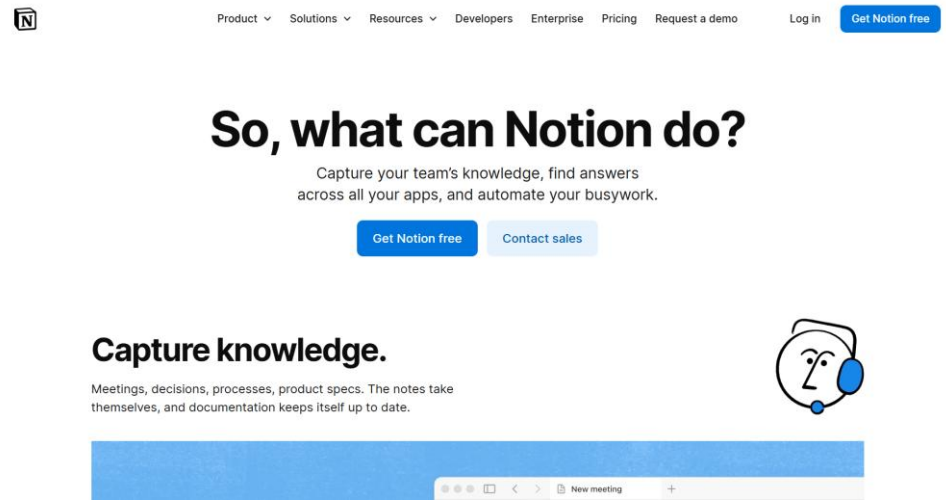


Figure 2.1 Notion Website Home Page

Notion is a cloud-based productivity app that integrates various features—such as note-taking, project management, databases, wikis, calendars, and collaboration—into a single platform. The app was developed to help individuals and teams organize information and boost productivity. In an educational context, Notion is widely used by students to take class notes, manage study schedules, handle assignments, and compile academic documents.

Some of the key features provided by Notion include:

1. Rich text note editor
2. Database and table management
3. Kanban board
4. Calendar view
5. Task management
6. File and media embedding
7. Collaboration workspace
8. Template gallery
9. AI writing assistant (Notion AI)

Notion has an intuitive and flexible interface that allows users to customize the structure of their workspace to suit their needs. The integration of various features into a single platform reduces the need to use many different apps. Additionally, real-time collaboration support makes it easier for students to work on group assignments and share information.

Although it offers a comprehensive set of features, Notion is not specifically designed as a personal learning system. Its AI features focus more on text creation and editing than on guiding the learning process. Furthermore, Notion does not yet provide a learning roadmap, structured progress tracking, or personalized learning resource recommendations based on users' learning goals.

2.2.2 Obsidian

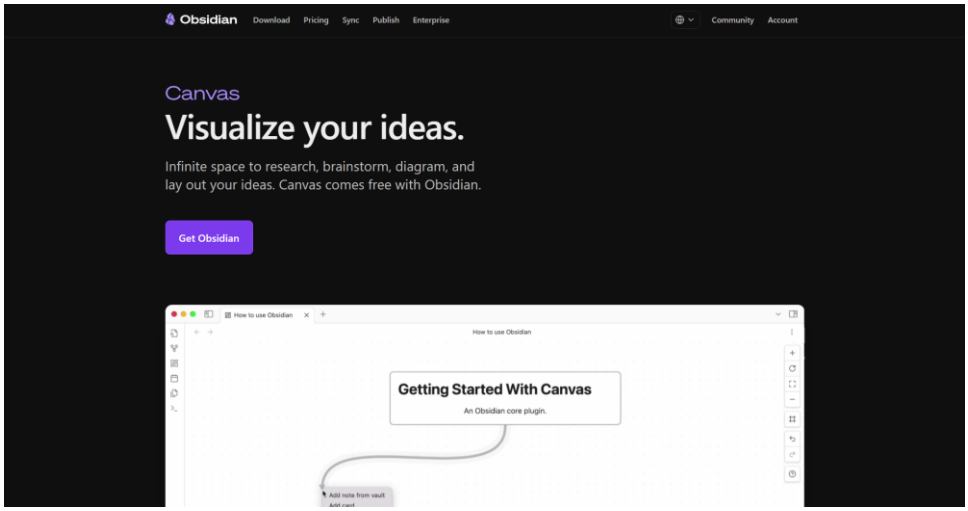


Figure 2.2 Obsidian Website Home Page

Obsidian is a Markdown-based personal knowledge management app that allows users to build a knowledge network through the concepts of linked notes and a knowledge graph. This app is widely used by researchers, students, and professionals to apply the "Second Brain" method in managing personal knowledge.

Obsidian's key features include:

1. Markdown editor
2. Internal note linking
3. Graph view
4. Backlinks
5. Tags and folders
6. Local file storage
7. Community plugins
8. Canvas visualization

9. Daily notes

Obsidian excels at connecting various knowledge concepts, making it easier for users to understand the relationships between topics. Local data storage provides full control over user data and allows for offline use. Plugin support also enables users to add various additional functions as needed.

Obsidian has a relatively steep learning curve for new users because many features require manual configuration. Additionally, the app focuses more on note management than on the learning process. Obsidian does not natively provide features such as learning goals, learning roadmaps, progress tracking, or AI-based learning recommendations.

2.2.3 Moodle

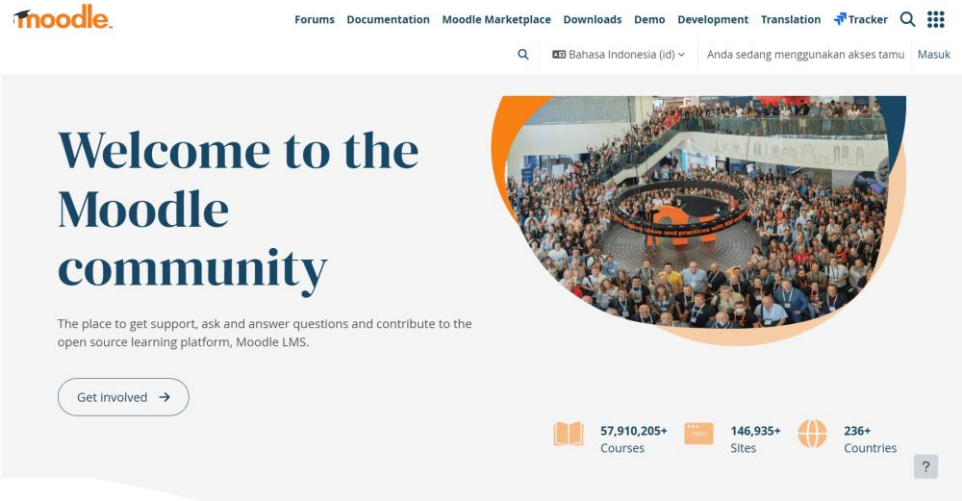


Figure 2.3 Moodle Website Home Page

Moodle is an open-source Learning Management System (LMS) widely used by educational institutions around the world to support online learning. Moodle provides a variety of tools for instructors and students to manage course materials, assignments, quizzes, discussion forums, and learning assessments all on a single platform.

Moodle's key features include:

1. Course management
2. Assignment submission
3. Quizzes and exams
4. Discussion forums
5. Gradebook

6. Learning resources
7. Attendance management
8. User management
9. Learning analytics

Moodle provides a comprehensive learning environment to support formal teaching and learning processes. The system supports various academic activities, ranging from distributing course materials and collecting assignments to grading and facilitating communication between instructors and students. Additionally, because it is open source, Moodle can be customized to meet the needs of the institution.

Moodle is more focused on classroom management than on managing students' personal knowledge. The system does not provide personal knowledge management features such as personal note-taking, knowledge graphs, or learning reference management. Furthermore, AI-based learning personalization features are still limited and generally require additional plugins.

2.2.4 Comparison with Proposed System

The following is a comparison of features between the proposed system and several existing platforms.

Feature	Notion	Obsidian	Moodle	Proposed System
Note Taking	✓	✓	Limited	✓
Knowledge Management	✓	✓	Limited	✓
Learning Goal Management	Limited	✗	Limited	✓
Learning Roadmap	✗	✗	✗	✓
Progress Tracking	Limited	✗	✓	✓
AI Search	Limited	Plugin	✗	✓
AI Summarization	Limited	Plugin	✗	✓
Learning Recommendation	✗	✗	✗	✓
Resource Management	✓	✓	✓	✓

Personal Dashboard	Limited	Plugin	Limited	✓

2.1 Comparison Table

Based on this comparison, it can be concluded that there is currently no platform that specifically integrates personal learning, personal knowledge management, self-directed learning, and artificial intelligence into a single system designed for students. Therefore, this study proposes the development of a system capable of combining all these functions to support the learning process more effectively and in a more personalized manner.

2.3 Conclusion

This chapter has discussed the theories that form the foundation of this research, including digital learning, personalized learning, knowledge management, personal knowledge management, self-directed learning, artificial intelligence, and the proposed system concept. In addition, a review was conducted of three widely used platforms—Notion, Obsidian, and Moodle—to identify their strengths and limitations.

The results of the review indicate that while each platform excels in specific aspects, none has yet been able to comprehensively integrate personal learning management, knowledge management, progress tracking, and AI support into a single system. Therefore, this study proposes the development of a Personal Learning and Knowledge Management System that combines these various concepts to help students manage their learning processes and knowledge more effectively, in a structured manner, and sustainably.

CHAPTER 3
SYSTEM ANALYSIS

3.1 Introduction

his chapter presents the system analysis and design of ChiroMemo, a web-based Personal Learning and Knowledge Management System (PLKMS) developed to support university students in organizing and managing their learning activities more effectively. Before the implementation phase begins, it is essential to conduct a comprehensive analysis of the proposed system to ensure that every component, process, and data structure is carefully planned and integrated. A well-designed system not only improves the overall functionality and reliability of the application but also minimizes potential issues during development and future maintenance.

The system analysis focuses on identifying the functional requirements, user interactions, data structures, and workflows required to support the core features of ChiroMemo. These include user authentication, learning goal management, learning roadmap management, learning resource organization, personal note management, AI-powered functionalities, and learning progress monitoring. Each component is designed to work together as part of an integrated learning ecosystem that promotes effective knowledge management and self-directed learning.

Furthermore, this chapter describes the proposed system development process and presents the system models that serve as the foundation for implementation. These include the system requirements, process modeling using Data Flow Diagrams (DFD), logical modeling through system flowcharts and user activity flows, and data modeling using the Entity Relationship Diagram (ERD). Together, these analyses provide a clear blueprint for developing ChiroMemo into a functional, maintainable, and user-friendly web application.

3.2 System Development Process

This section will explain the development process applied to the self-directed learning and knowledge management system (ChiroMemo). This development process must be carried out systematically to ensure an efficient system that meets the project's objectives. This project adopts an Object-Oriented Analysis and Design (OOAD) approach along with the Agile software development methodology. OOAD provides a structured approach to analyzing user requirements and designing system architecture, while Agile supports iterative development, allowing the system to be developed, tested, and improved in stages. The overall development process consists of project planning, requirements analysis, system design, implementation, testing, and deployment.

3.2.1 Project Planning and Oversight

In this development, the proposed system will utilize the Agile software development methodology combined with an Object-Oriented Analysis and Design (OOAD) approach. The Agile methodology was chosen because it enables iterative and incremental development, allowing each system module to be developed, evaluated, and refined throughout the project. This approach provides flexibility in accommodating improvements during the development process while ensuring that the final system meets the identified user requirements.

Meanwhile, OOAD is used to analyze system requirements and design the software architecture using object-oriented principles. This approach facilitates the identification of system entities, the relationships between them, and their interactions, resulting in a system design that is modular, easy to maintain, and scalable.

3.2.2 System Requirement Analysis

This system requirements analysis will be conducted to identify system requirements, functional requirements, and non-functional requirements. This analysis ensures that the developed system meets students' needs by providing an integrated platform for managing learning activities, organizing knowledge, and supporting self-directed learning.

System Requirements

1. **User Registration**

   Input
   - Full Name
   - Email Address
   - Password
   - Confirm Password

   Process
   - Validate the entered information.
   - Verify that the email address is not already registered.
   - Encrypt the password before storing it.
   - Save the user information into the database.

   Output
   - Registration successful message.
   - Redirect the user to the Login page.

2. **User Login**

Input

- Email Address
- Password

Process

- Validate the login credentials.
- Authenticate the user using the stored account information.
- Create a user session upon successful authentication.

Output

- Redirect the user to the Dashboard.
- Display an error message if authentication fails.

3. **Learning Goal Management**

Input

- Goal Title
- Goal Description
- Target Completion Date

Process

- Validate the entered information.
- Save the learning goal into the database.
- Associate the goal with the authenticated user.

Output

- Learning goal successfully created.
- Display the updated list of learning goals.

4. **Learning Roadmap Management**

Input

- Roadmap Title
- Roadmap Description
- Roadmap Steps
- Target Completion Date

Process

- Validate the roadmap information.
- Create roadmap milestones.

- Store the roadmap and its associated steps in the database.

Output

- Learning roadmap successfully created.
- Display the roadmap progress.

5. Learning Resource Management

Input

- Resource Title
- Resource Type (PDF, Video, Website, Book, Course)
- Resource URL or Uploaded File
- Category

Process

- Validate the resource information.
- Store the learning resource in the database.
- Associate the resource with the current user.

Output

- Learning resource successfully added.
- Display the updated resource library.

6. Personal Note Management

Input

- Note Title
- Note Content
- Related Learning Resource (Optional)

Process

- Validate the note information.
- Save the note into the database.
- Link the note to the selected learning resource if provided.

Output

- Note successfully created or updated.
- Display the updated list of notes.

7. AI-Powered Intelligent Search

Input

- Search Query
- Process
- Receive the user's query.
- Send the query to the AI service.
- Retrieve relevant information based on the user's learning resources and query.

Output

- Display AI-generated search results.
- Provide relevant learning resources and suggestions.

8. AI-Powered Document Summarization

Input

- Learning Document (PDF or Text)

Process

- Upload the document.
- Send the document to the AI service.
- Generate a concise summary.
- Store the generated summary in the database.

Output

- Display the summarized content.
- Save the summary for future reference.

9. Personalized Learning Recommendation

Input

- User learning goals
- Learning progress
- Learning resources
- Learning activity history

Process

- Analyze the user's learning goals and activity data.
- Identify relevant learning topics and resources.

- Send the relevant information to the AI service.
- Generate personalized learning recommendations.

Output

- Display recommended learning resources.
- Display recommended topics or learning activities.
- Allow users to save relevant recommendations for future reference.

Functional Requirements

ID	Functional Requirement
FR-01	Users shall be able to register a new account.
FR-02	Users shall be able to log in securely.
FR-03	Users shall be able to create, edit, and delete learning goals.
FR-04	Users shall be able to create and manage learning roadmaps.
FR-05	Users shall be able to organize learning resources such as links, documents, and references.
FR-06	Users shall be able to create and manage personal notes.
FR-07	Users shall be able to monitor learning progress through a dashboard.
FR-08	Users shall be able to perform AI-powered intelligent searches.
FR-09	Users shall be able to generate summaries of learning documents using Artificial Intelligence.
FR-10	Users shall receive personalized learning recommendations based on their learning activities.

3.1 Functional Requirements Table

Non-Functional Requirments

Category	Requirement

Performance	The system should provide responses within an acceptable time under normal usage.
Security	User passwords shall be encrypted and authentication shall be required before accessing personal data.
Usability	The user interface should be simple, intuitive, and easy to navigate.
Compatibility	The system shall support modern web browsers on desktop and mobile devices.
Availability	The system shall be accessible through an internet connection at any time.
Maintainability	The system shall be developed using a modular architecture to facilitate future maintenance and enhancements.

3.2 Non-Functional Requirement Table

User Requirements

The proposed ChiroMemo system is designed based on the learning needs of university students who use multiple digital resources during their learning activities. The system is intended to provide a centralized platform where students can organize learning resources, manage personal notes, establish learning goals, create learning roadmaps, and monitor their learning progress.

Users are also expected to use AI-powered features to improve the efficiency of their learning activities. These features include intelligent search, document summarization, and personalized learning recommendations. Therefore, the system should provide an interface that is simple, intuitive, and accessible through modern web browsers.

The main user requirements are as follows:

1. Users should be able to create and manage their personal accounts.
2. Users should be able to create and manage learning goals.
3. Users should be able to create structured learning roadmaps.
4. Users should be able to organize and manage learning resources.
5. Users should be able to create, edit, and manage personal notes.
6. Users should be able to monitor their learning progress through the dashboard.

7. Users should be able to search for learning information using AI-powered intelligent search.
8. Users should be able to summarize learning documents using AI.
9. Users should receive personalized learning recommendations based on their learning activities and goals.

3.3 System Modeling

The system modeling section will discuss the system by presenting several models that are expected to help explain the workflow of the system currently under development.

3.3.1 Process Modeling

Process modeling describes how information flows throughout the proposed system and how different processes interact with users and data storage. It provides a clear understanding of the system workflow before implementation begins. The process modeling of ChiroMemo consists of a Context Diagram, Data Flow Diagram (DFD) Level 0, and DFD Level 1.

3.3.1.1 Context Diagram

The Context Diagram illustrates the overall interaction between the external entity and the proposed ChiroMemo system. It provides a high-level overview of the system boundaries without showing the internal processes. In this project, the university student is the primary external entity who interacts with the system to perform learning activities.

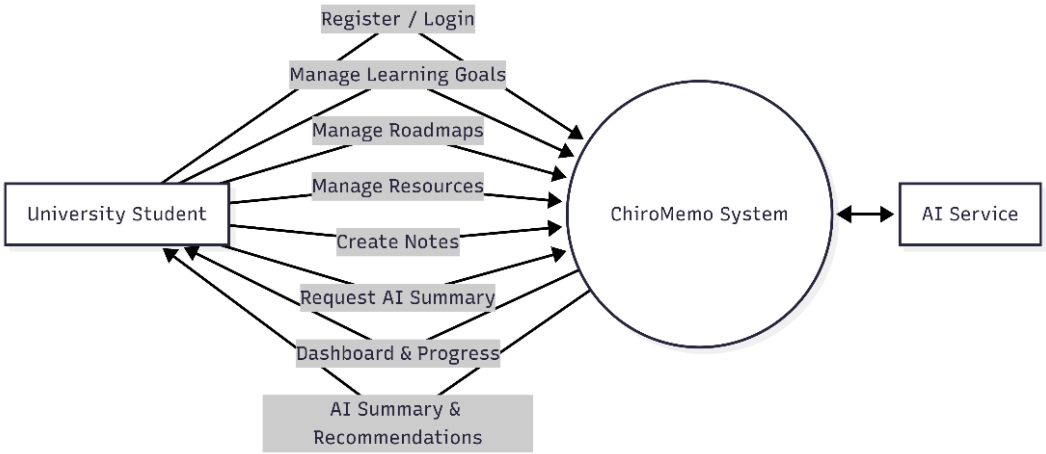


Figure 3.1 Context Diagram for ChiroMemo System

Figure 3.1 shows that the University Student interacts directly with the ChiroMemo system. The user can register an account, log in, manage learning goals, create learning roadmaps, organize learning resources, write personal notes, and request AI-powered services such as document summarization and intelligent search. The system processes these requests and returns outputs

such as learning progress, AI-generated summaries, personalized recommendations, and dashboard information. For AI-related features, the system communicates with an external AI service to generate the requested results.

3.3.1.2 Data Flow Diagram Level 0

The DFD Level 0 illustrates the major processes within the proposed system. The system is divided into several functional modules, including User Management, Learning Goal Management, Learning Roadmap Management, Resource Management, Note Management, AI Services, and Dashboard. Each process exchanges data with the central database to ensure data consistency and efficient information management.

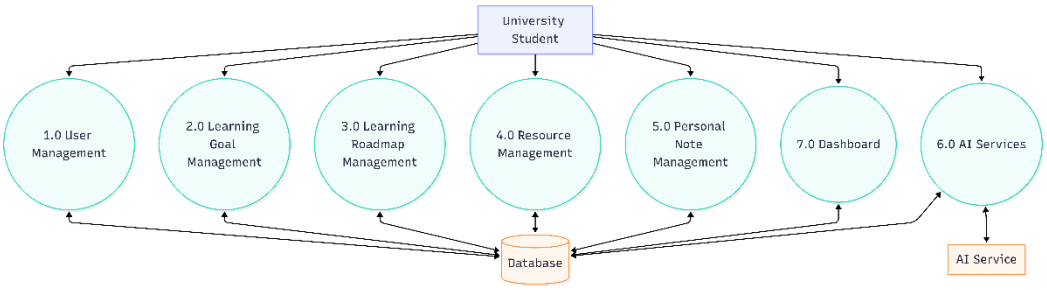


Figure 3.2 DFD Level 0 for ChiroMemo System

As illustrated in Figure 3.2, the proposed system consists of seven main processes:

1. User Management handles user registration, authentication, profile updates, and logout.
2. Learning Goal Management allows users to create, update, delete, and monitor their learning goals.
3. Learning Roadmap Management enables users to organize learning milestones and monitor roadmap progress.
4. Learning Resource Management manages learning materials such as websites, documents, and videos.
5. Personal Note Management allows users to create and organize personal learning notes.
6. AI Services provide intelligent search, document summarization, and personalized learning recommendations by communicating with the external AI service.
7. Dashboard retrieves information from multiple modules to present learning progress and activity summaries.

3.3.1.3 Data Flow Diagram Level 1

The DFD Level 1 provides a more detailed view of selected processes in the system. Each module is decomposed into smaller processes that describe how user requests are handled. For example, the Learning Goal Management module consists of creating, updating, deleting, and monitoring learning goals, while the AI Services module includes intelligent search, document summarization, and personalized recommendation functionalities.

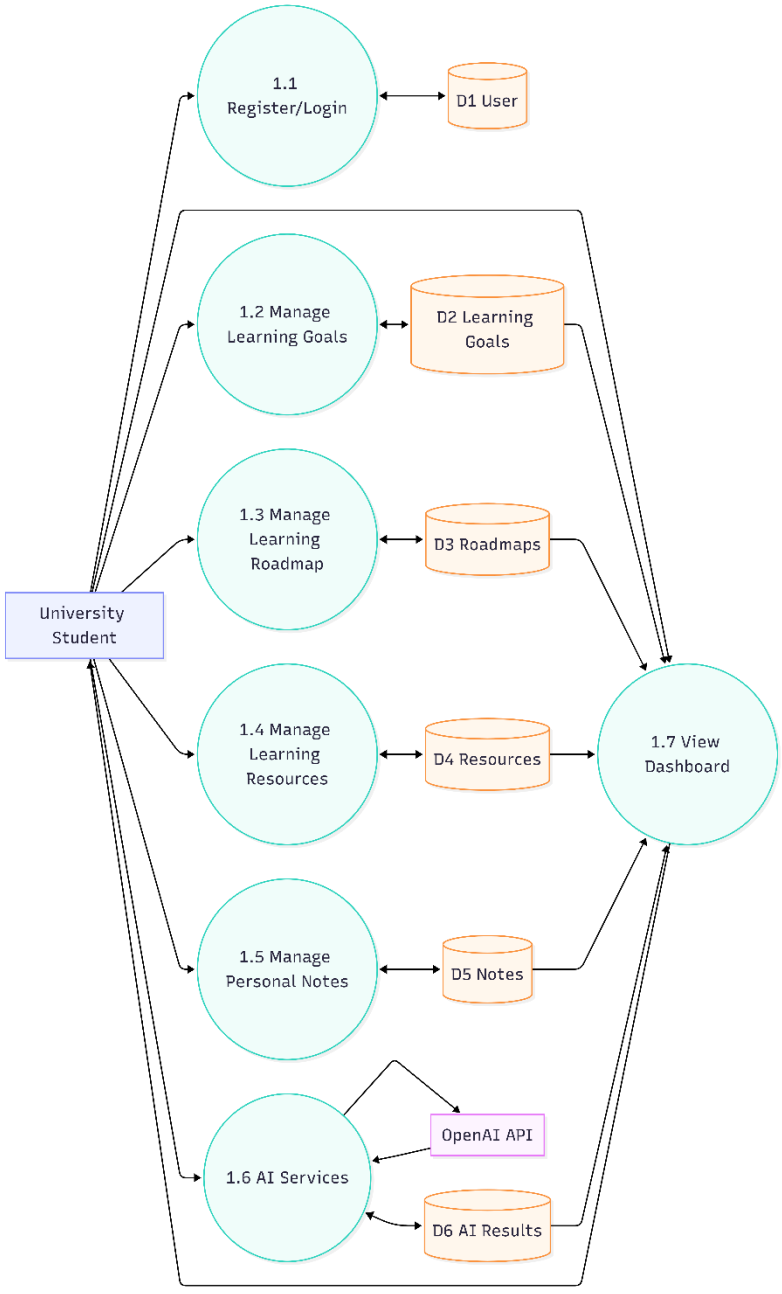


Figure 3.3 DFD Level 1 for ChiroMemo System

Figure 3.3 illustrates the detailed workflow of the proposed ChiroMemo system. The interaction begins when the university student accesses the system through the User Management process for registration or login. After successful authentication, the user can manage learning goals, create learning roadmaps, organize learning resources, and maintain personal notes. The user may also access AI-powered services to perform intelligent searches, summarize learning documents, and receive personalized learning recommendations. All learning data are stored in their respective databases, while AI-generated results are saved for future reference. Finally, the Dashboard retrieves data from the various modules to present an overview of the user's learning progress and recent activities.

3.3.2 Logical Modeling

The logical modeling phase will describe the structure of the logic flow to be implemented in ChiroMemo. This section will explain how users will interact with the system, from the moment they log in until they log out.

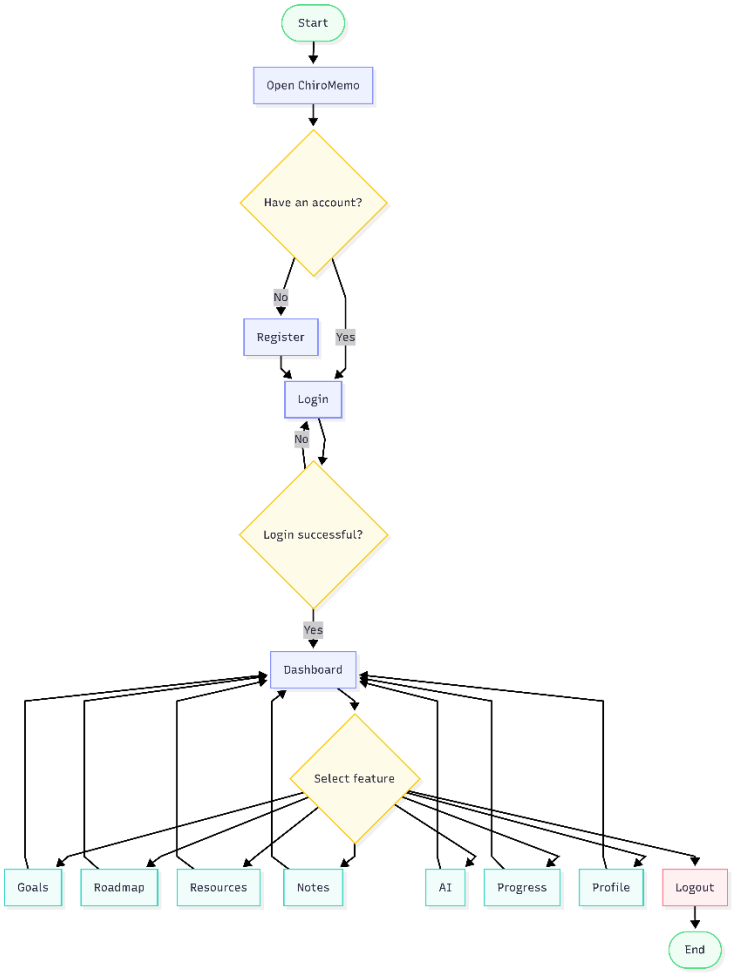


Figure 3.4 Flowchart for ChiroMemo System

Figure 3.4 shows a flowchart of the ChiroMemo system. This flowchart illustrates how users will interact with the system, from authentication, setting goals, creating a roadmap, saving notes, viewing progress, and interacting with AI.

3.3.3 Data Modelling

Data modeling describes the structure of the database used in the ChiroMemo system. The purpose is to the main entities, their attributes, and the relationships between them. The Entity Relationship Diagram (ERD) is used to illustrate how data is organized and connected to support the system's functionality.

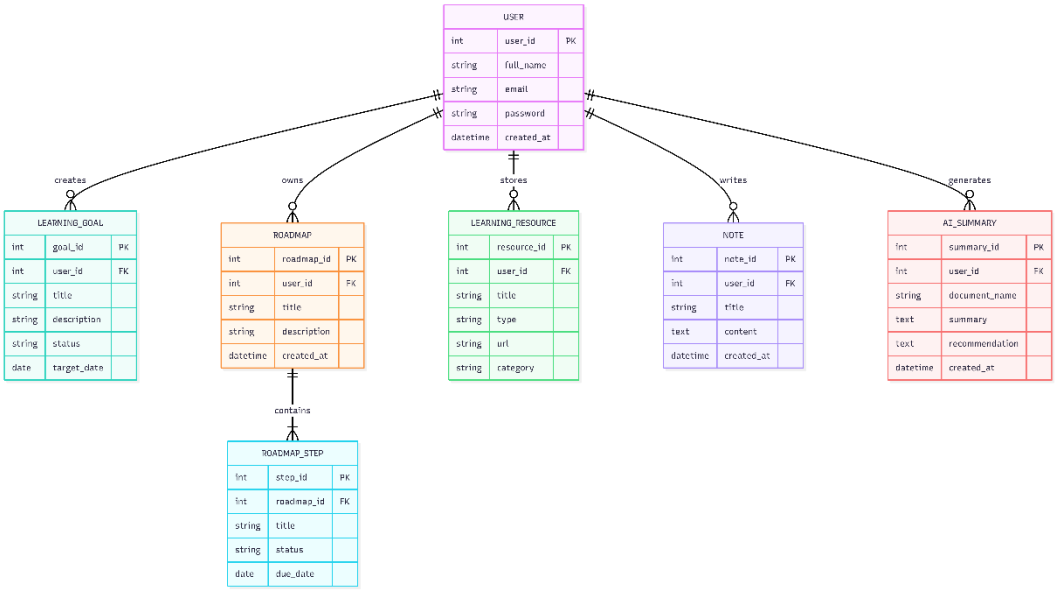


Figure 3.5 ERD for ChiroMemo System

Figure 3.5 illustrates an Entity-Relationship Diagram (ERD) of the ChiroMemo system. This ERD shows the relationships between users, goals, roadmaps, resources, notes, and summaries. The ERD serves as the database design that supports the main functions of the proposed system.

CHAPTER 4
SYSTEM DESIGN

4.1 Introduction

This chapter presents the design of the ChiroMemo application. The system design focuses on developing a simple, user-friendly, and visually organized interface to provide users with a comfortable learning experience. The proposed interface designs are presented using low-fidelity wireframes to illustrate the structure, layout, and main components of each system interface.

The designs cover the main functionalities of ChiroMemo, including user registration, user login, dashboard, learning goal management, learning roadmap management, learning resource management, personal note management, AI-powered intelligent search, and AI-powered document summarization.

4.2 Design Forms and Reports

4.2.1 Register Page

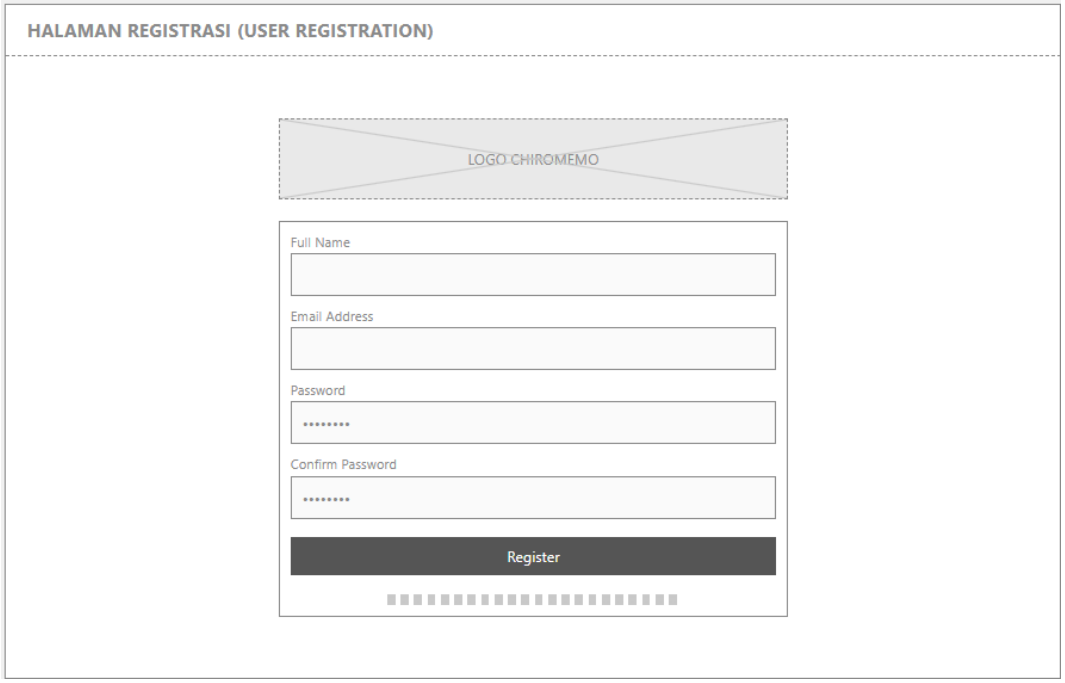


Figure 4.1 Register Page Wireframe ChiroMemo System

The Registration Page is designed to allow new users to create a ChiroMemo account before accessing the system. This page collects the basic information required to create a user account while ensuring that the registration process remains simple and user-friendly.

The interface consists of four input fields, namely Full Name, Email Address, Password, and Confirm Password, followed by a Register button to submit the registration request. The system validates the entered information before creating the account. If the registration is successful, the user will be redirected to the Login Page to access the application.

4.2.2 Login Page

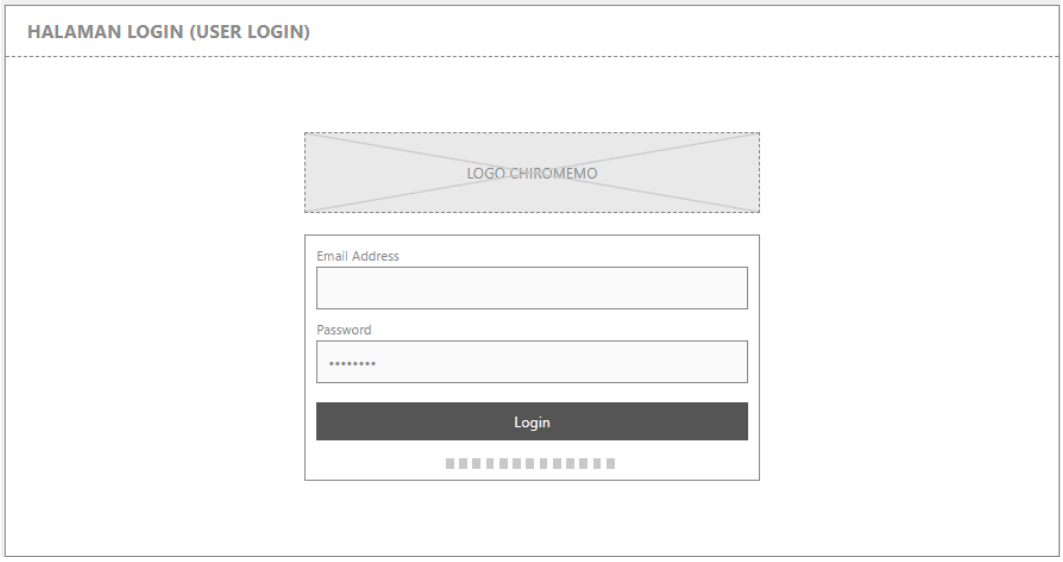


Figure 4.2 Login Page Wireframe ChiroMemo System

The Login Page enables registered users to access ChiroMemo securely using their existing credentials. This page serves as the authentication gateway before users can interact with the application's learning management features.

The interface includes two input fields, Email Address and Password, together with a Login button. The system verifies the provided credentials against the user database. Upon successful authentication, users are redirected to the Dashboard, while invalid credentials will result in an appropriate error message.

4.2.3 Dashboard

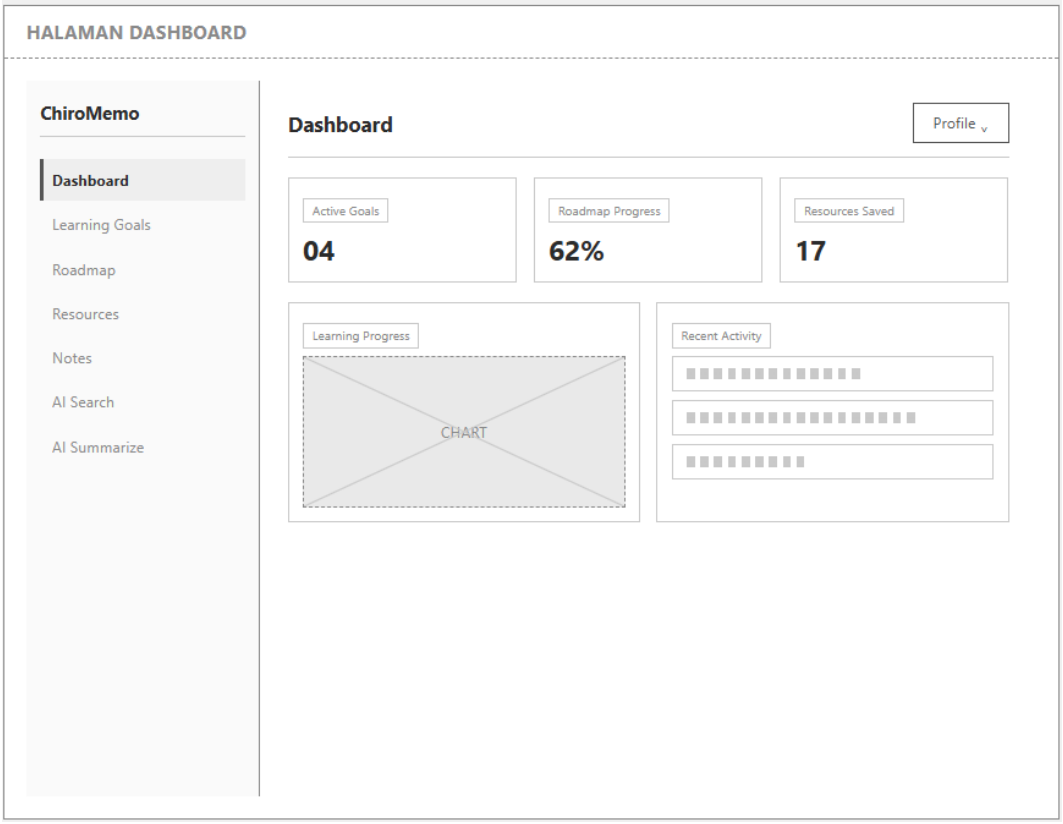


Figure 4.3 Dashboard Page Wireframe ChiroMemo System

The Dashboard serves as the main interface of ChiroMemo, providing users with an overview of their learning activities and progress. It is designed to present important information in a clear and organized manner, allowing users to monitor their learning performance efficiently.

The dashboard displays several summary components, including active learning goals, roadmap progress, saved learning resources, learning progress charts, and recent learning activities. These components provide users with quick access to important information and help them identify their current learning status.

4.2.4 Learning Goal Management

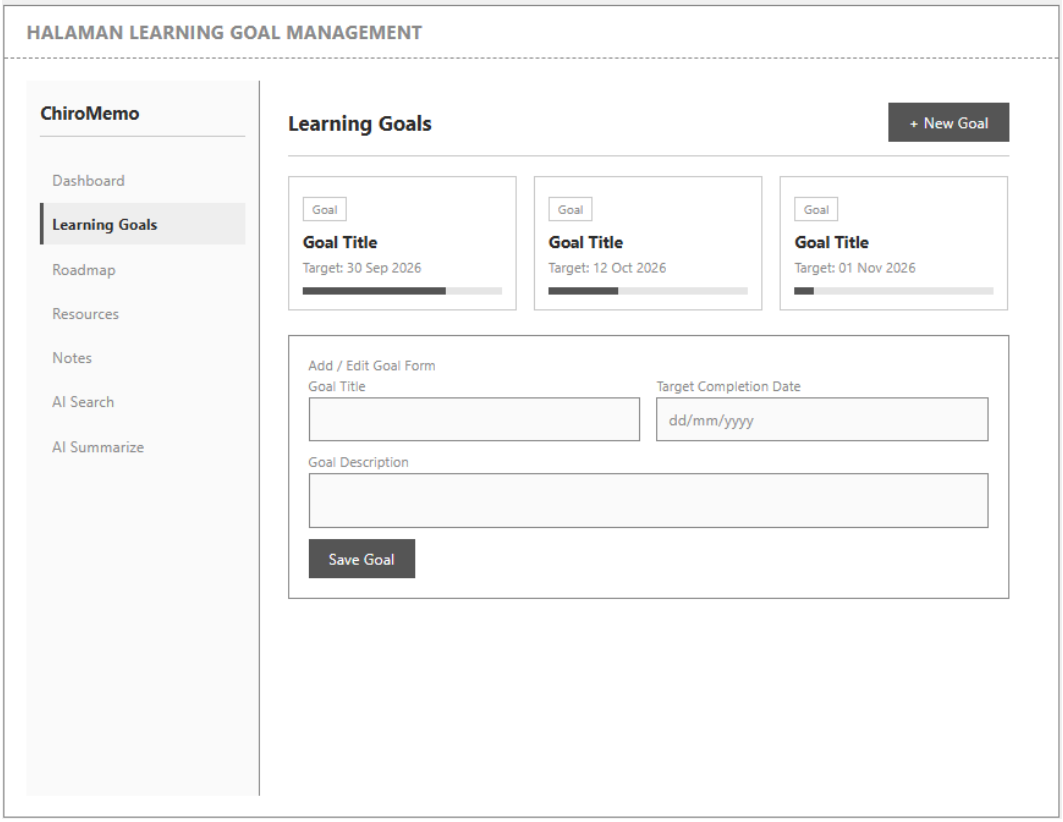


Figure 4.4 Learning Goal Management Page Wireframe ChiroMemo System

The Learning Goal Management page allows users to create, manage, and monitor their personal learning objectives. This feature encourages users to define clear learning targets and track their achievement throughout the learning process.

The interface presents learning goals in the form of cards, each displaying the goal title, completion status, and progress bar. Users can add new goals, edit existing goals, update their progress, or remove completed or unnecessary goals.

4.2.5 Learning Roadmap Management

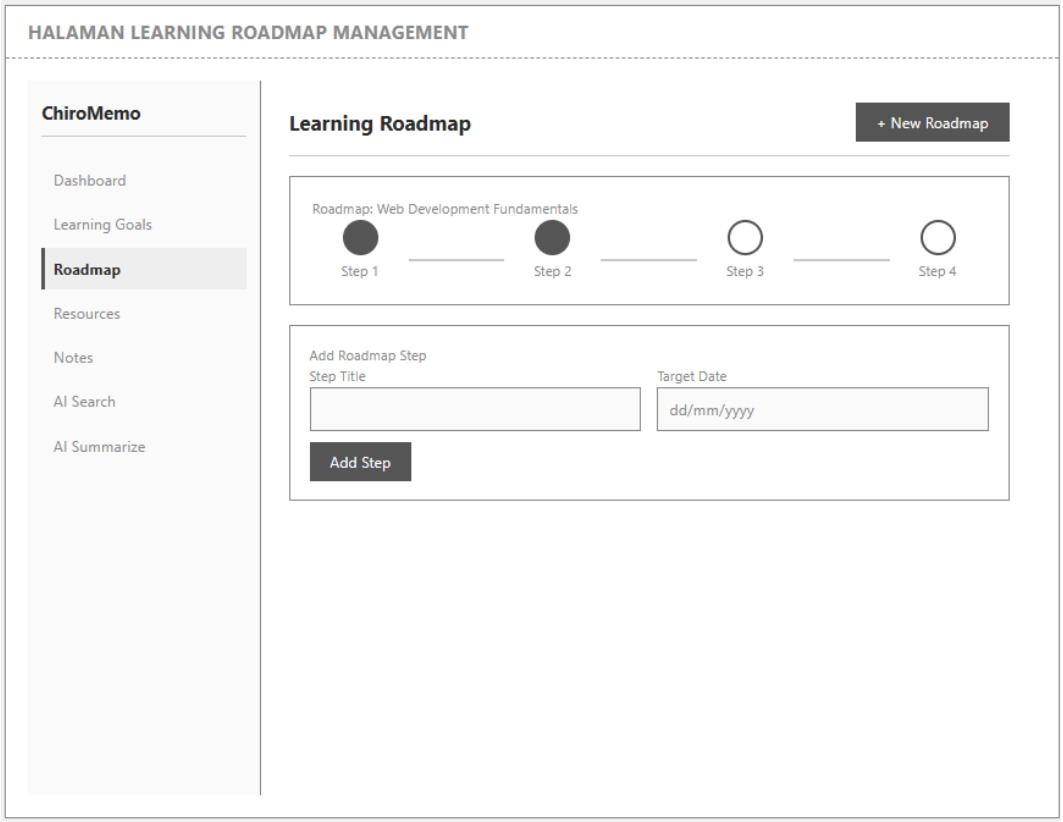


Figure 4.5 Learning Roadmap Management Page Wireframe ChiroMemo System

The Learning Roadmap Management page helps users organize their learning journey into structured milestones and sequential learning steps. This feature supports self-directed learning by allowing users to visualize their learning path.

The page displays the roadmap as a sequence of learning milestones together with their completion status. Users can create new roadmaps, add learning steps, edit existing milestones, and update their progress as they complete each stage.

4.2.6 Learning Resource Management

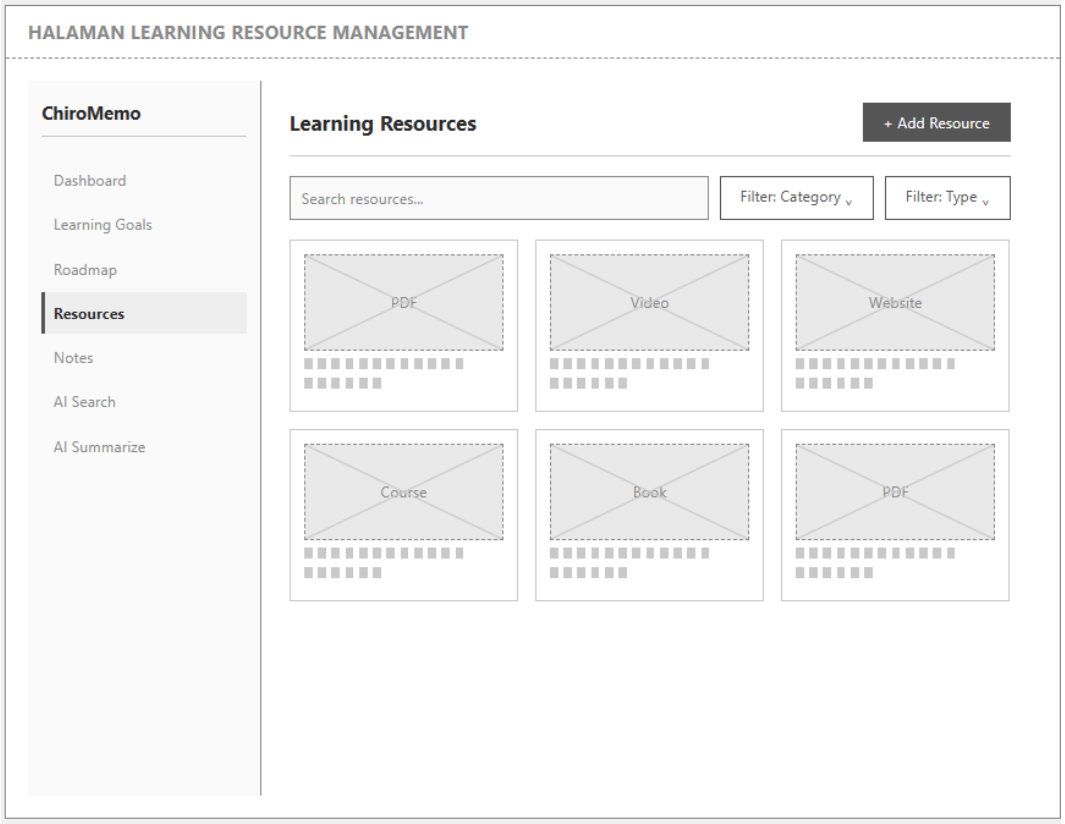


Figure 4.6 Learning Resource Management Page Wireframe ChiroMemo System

The Learning Resource Management page is designed to organize learning materials collected from various sources into a centralized repository. This enables users to manage their study materials more efficiently.

Learning resources are displayed in a grid layout, where each resource contains its title, category, and type such as PDF, Video, Website, Course, or Book. The page also provides search and filter functions to help users quickly locate specific learning resources.

4.2.7 Personal Note Management

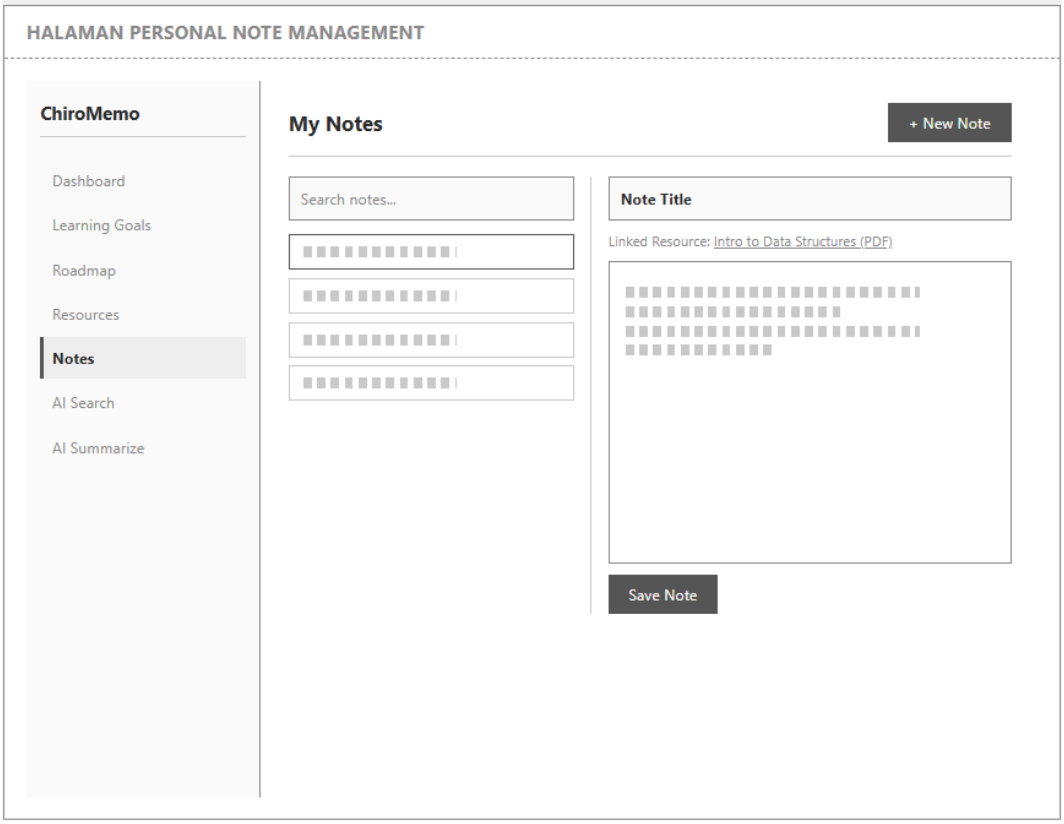


Figure 4.7 Personal Note Management Page Wireframe ChiroMemo System

The Personal Note Management page enables users to create and organize personal notes while studying. This feature allows users to record important information, summarize learning materials, and keep knowledge in a structured format.

The interface consists of a note list displayed on the left side and a note editor on the right side. Users can create, edit, delete, and organize notes while linking them to related learning resources for easier referen

4.2.8 AI Intelligent Search

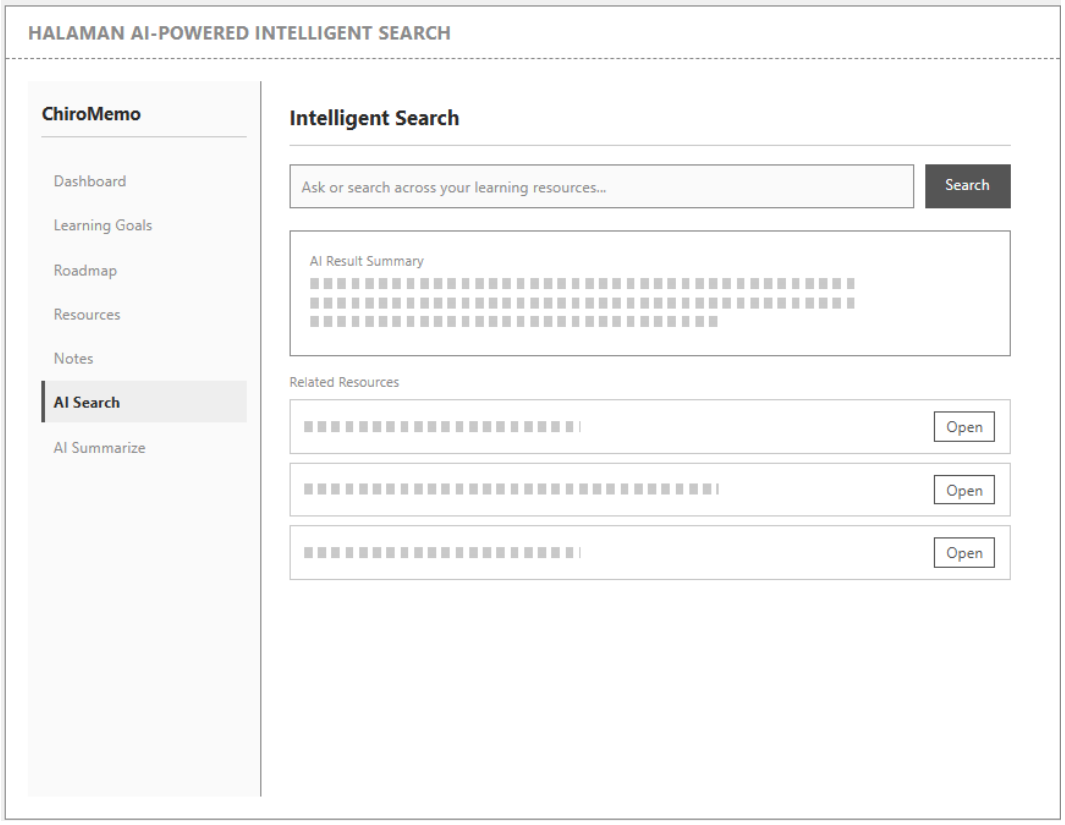


Figure 4.8 AI Intelligent Search Page Wireframe ChiroMemo System

The AI-Powered Intelligent Search page allows users to search for learning information using natural language queries. By utilizing artificial intelligence, the system provides more relevant search results compared to conventional keyword-based searches.

The page contains a search bar where users can enter their questions or keywords. After processing the query, the system displays an AI-generated response together with related learning resources that may assist users in understanding the requested topic.

4.2.9 AI Document Summarization

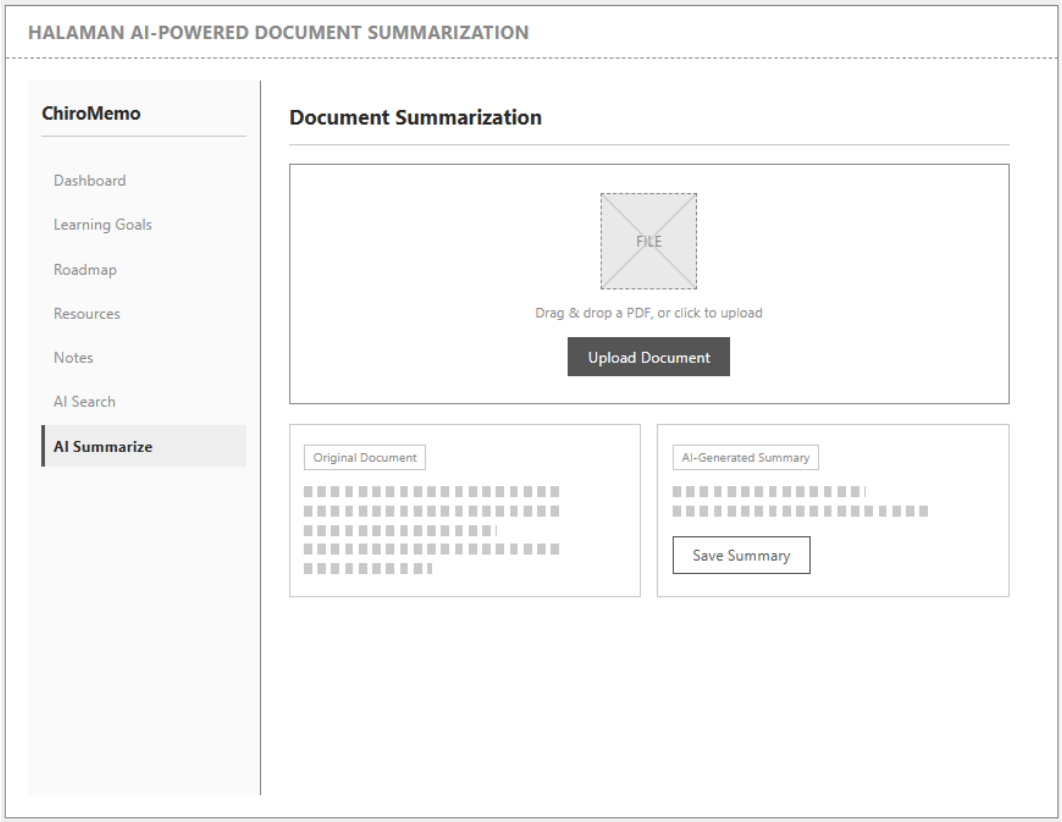


Figure 4.9 AI Document Summarization Page Wireframe ChiroMemo System

The AI-Powered Document Summarization page enables users to upload learning documents and obtain concise summaries generated by artificial intelligence. This feature helps students quickly understand lengthy learning materials without reading the entire document.

The interface includes a document upload area, followed by two display panels showing the original document and the AI-generated summary. Users can review the summarized content and save it for future reference.

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